Step-by-Step Methods

Laws of Exponents - Scientific Notation - Real Numbers

FIRST: LEARN THE 5 LAWS

If you see...	Do this	Example
Multiply (same letter)	**ADD** the exponents	$a^2 \times a^3 = a^5$
Divide (same letter)	**SUBTRACT** (top - bottom)	$a^7 / a^2 = a^5$
Bracket with a power	**MULTIPLY** the exponents	$(a^2)^3 = a^6$
Negative exponent	**MOVE** it to the other floor and make it positive	$a^{-2} = 1 / a^2$
Zero exponent	The answer is **1**	$7^0 = 1$

TYPE 1 - FIND THE VALUE OF A NUMBER (like Q2 and Q4)

Step 1: If the exponent is **zero**, the answer is **1**. Careful: $-9^0 = -1$ (the minus is outside), but $(-9)^0 = 1$.
Step 2: If the exponent is **negative**, write 1 over the number with a **positive** exponent. If it is a fraction, **flip** the fraction instead.
Step 3: Calculate the number to the end. Do not stop at the exponent.

Examples

Negative exponent	$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$
Fraction: flip it	$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$
Zero exponent	$5^0 = 1$

TYPE 2 - SIMPLE SIMPLIFYING (like Q3)

Step 1: Look at the operation first: **multiply**, **divide** or **bracket**?
Step 2: Use the matching law from the table.

Example

Multiply -> add the exponents	$b^5 \times b^2 = b^{5+2} = b^7$

TYPE 3 - BIG QUESTIONS (like Q5 - Q8 and Q10)

Step 1: Open the brackets. The power outside goes to **everything inside, numbers too**: $(ab)^3 = a^3b^3$ and $(3a^2)^2 = 9a^4$.
Step 2: Dividing two fractions? Keep - Change - Flip: keep the first fraction, change / to x, flip the second.
Step 3: Collect everything on top together and everything on the bottom together. Add the exponents of the same letter on each floor: $a^2 \times a^3 = a^5$.
Step 4: Compare each letter on top with the same letter on the bottom. Subtract: top exponent - bottom exponent. Divide the ordinary numbers.
Step 5: Clean up. Any negative exponent moves to the other floor and becomes positive. Any letter with exponent 0 disappears (it equals 1).

Full example

Question	$\frac{a^3b}{c^2} \times \frac{(bc)^2}{a}$
Step 1: open the bracket	$(bc)^2 = b^2c^2$
Step 3: top and bottom	$\frac{a^3 b^3 c^2}{c^2 a}$
Step 4: a: 3 - 1 = 2, c: 2 - 2 = 0	$a^2 b^3 c^0$
Step 5: answer	a^2b^3

Never cross-cancel in an X shape. You can only cancel between the **top** and the **bottom**.

TYPE 4 - FIND THE MISTAKE (like Q9)

Step 1: Look at **each step on its own** and ask: which law was used here?
Step 2: Solve it yourself correctly and compare your result with the student's.
Step 3: Write the mistake in one sentence: "The student did ..., but the correct answer is ..., because the law says ...".
Step 4: At the end, **write the full correct solution** with all the steps.

Where mistakes usually hide:
• The number inside the bracket was **not raised to the power**: $(2a^3)^2 = \mathbf{4}a^6$, not $2a^6$.
• Subtracting a **negative**: 5 - (-3) = 5 + 3 = **8**, not 2.

TYPE 5 - SCIENTIFIC NOTATION

Form: a x 10^n, where a is from 1 up to (but not including) 10.

Step 1: Move the decimal point until only **one digit** (not zero) is in front of it.
Step 2: Count the places you moved: that is the exponent.
Step 3: Big number -> **positive** exponent. **Small** number (0.00...) -> **negative** exponent.
Step 4: Multiply: multiply the numbers, ADD the exponents. **Divide:** divide the numbers, SUBTRACT the exponents.
Step 5: If the front number is 10 or more, move the point one place left and add 1 to the exponent.

Examples

Big number	$45\,000 = 4.5 \times 10^4$
Small number	$0.0072 = 7.2 \times 10^{-3}$
Multiply, then fix	$(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 = 2 \times 10^6$

TYPE 6 - REAL NUMBERS & SQUARE ROOTS

Step 1: Can the number be written as a **fraction**? Yes -> **rational**. No -> **irrational**.
Step 2: Decimal that **stops** or **repeats** -> rational. Never stops and never repeats (pi) -> irrational.
Step 3: √ of a perfect square (√25 = 5) -> rational. √ of any other number (√7) -> irrational.
Step 4: To estimate a square root, find the two perfect squares it sits between: √30 is between √25 = 5 and √36 = 6.

Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
Sets: Natural (1, 2, 3...) -> Whole (adds 0) -> Integers (adds negatives) -> Rational (fractions) -> Real (rational + irrational)

FINAL CHECK BEFORE YOU MOVE ON

- Are there **no brackets** left?
- Are there **no negative exponents** left?
- Did you **finish the calculation** (e.g. $1/5^3 = 1/125$)?
- Did you write **every step**? Steps earn marks even if the final answer is wrong.

Study tips: keep this sheet next to you while solving. Say each step out loud. Solve one big question at a time, slowly.